

Computing Squares of Uncommon Numbers Efficiently

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1 Introduction

During the solving of questions, it is definitely annoying to calculate squares of random numbers, say 67^2 or 32^2 or 103^2 or 94^2 . Definitely, one can multiply these to get the results, but there is another way that makes these calculations easier.

NOTE: It is recommended to implement them in daily solving and over time, such squares will become a piece of cake to calculate.

2 Squares of Numbers ending in '5'

Take an arbitrary natural number ending in '5' (5 is in the units place): $(10x + 5)^2$ where x is an integer — it can be any number, even a multiple of 100, giving rise to larger numbers, say 115^2 being written as $(10(11) + 5)^2$ where $x = 11$.

$$\begin{aligned}(10x + 5)^2 &= (10x)^2 + 2(10x)(5) + (5)^2 \quad \text{using the square of a sum} \\ &= 100x^2 + 100x + 25 \\ &= 100x(x + 1) + 25\end{aligned}$$

$100x(x + 1)$ is a number always ending in two zeroes, so the units and tens places are always 0 and 0, respectively. Now we're adding 25 to this, so the two zeroes in the units and tens places will be replaced by 2 and 5, respectively.

This indicates something beautiful. If we want to calculate the square of $(10x + 5)^2$, we can take x and multiply it by $x + 1$, and append the result with 25 to get the final square.

Some examples:

65^2 : $6 \times 7 = 42$, and we append 42 with 25 to get the **square of 65** as **4225**.

This will certainly work with other numbers because we have even established a proof.

Summary

1. Consider the number without 5 in the units place, e.g. for 105^2 , your number is 10.
2. Multiply the number from Step 1 by the succeeding number: $10 \cdot 11 = 110$.
3. Append the result from Step 2 with 25. We obtain 11025, the correct square of 105.

3 Squares of Uncommon Numbers

The approach is simple. We can easily calculate squares of numbers like $50^2, 100^2, 40^2$, and we will use them to calculate squares of random numbers like 53 or say 42.

Prerequisite: We know what x^2 is.

We need to calculate $(x + 1)^2$

$$(x + 1)^2 = x^2 + 2x + 1$$

Observe that $2x + 1$ can be written as $x + (x + 1)$, so

$$(x + 1)^2 = x^2 + x + (x + 1)$$

Remarks

We know what the square of the preceding number is, and we can calculate the square of the successive number by adding the number preceding it and the number itself to the square of the number preceding it. **Mathematically**,

$$\begin{aligned} 51^2 &= 50^2 + 50 + 51 \\ &= 2500 + 101 = 2601 \end{aligned}$$

Now say you want to calculate 52^2 ; then you take 51^2 and add $51 + 52 = 103$ to it.

$$52^2 = 2601 + 51 + 52 = 2704$$

This can be further extended for other numbers. When you come across a number after the middle value of a set of 10 numbers in order, we know how to **calculate the square of numbers ending in 5**, so we can extend it to this scenario.

Example

$$\begin{aligned} 56^2 &= 55^2 + 55 + 56 \\ &= 3025 + 111 = 3136 \end{aligned}$$

where we calculated 55^2 using $(5 \cdot 6)\underline{25}$, giving 3025.

4 Suggestion

This can feel overwhelming at first to use, but over time with daily usage, it will become almost inborn and you will be able to do this within seconds. Keep practicing!

Suggestion to the reader:

Go and try making some rules of your own for numbers like $(x+2)^2$ and $(x+3)^2$. Remembering rules for these is more than enough.

_____ **THE END** _____